Technical Supplement

Exact Certificates, Payback Diagnostics, and Experimental Protocols
for *Non-Monotone Service Placement and Reconfiguration Barriers in Edge Orchestration*

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Abstract

This companion retains instance specifications, algebraic derivations, full-state certificates, and experimental details supporting the conference manuscript. The manuscript is self-contained with respect to the system model, main structural claims, witness configurations, and conditional controller guarantee. Versioned research code, synthetic datasets, certificates, and this supplement are publicly available at `https://github.com/jozsefbirotmit/edge-orch-reprod`. The original research artifact (v1.0.0) is archived at `https://doi.org/10.5281/zenodo.24200109`. This revised supplement will belong to a subsequent artifact release, with a separate version DOI when Zenodo archiving is complete.

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1 Model details and the traffic transformation rule

There are n services, one traffic-transforming processing stage per service, and candidate processing tiers R, M, C along a common *logical* two-link route. A service's input traffic b_i becomes $\eta_i b_i$ after processing. The choice of tier affects aggregate carried traffic even though all services still experience the same two link-delay terms. The common logical route is a modelling abstraction; it does not imply shorter geographical paths or local-breakout routes when processing occurs closer to the RAN. For placement $\pi \in \{R, M, C\}^n$,

Table 1: Per-service contribution to shared transport-link loads

Stage tier	R–M load	M–C load
R	$\eta_i b_i$	$\eta_i b_i$
M	b_i	$\eta_i b_i$
C	b_i	b_i

write

$$q_t^\pi = \sum_{i:\pi_i=t} c_i,$$
$$L_1^\pi = \sum_i b_i[\eta_i \mathbf{1}\{\pi_i = R\} + \mathbf{1}\{\pi_i \neq R\}],$$
$$L_2^\pi = \sum_i b_i[\eta_i \mathbf{1}\{\pi_i \neq C\} + \mathbf{1}\{\pi_i = C\}].$$

The congestion surrogates and worst normalized service cost are

$$d_e(L, B_e) = \delta_e + a_e L/(B_e - L),$$
$$P_i^\pi(H) = p^0_{i,\pi_i} \frac{H_{\pi_i}}{H_{\pi_i} - q^\pi_{\pi_i}},$$
$$J_\pi(H) = \max_i \frac{r_i + \sum_e d_e(L_e^\pi, B_e) + P_i^\pi(H)}{\tau_i}.$$

Hard feasibility means $L_e^\pi < B_e$ and $q_t^\pi < H_t$ for occupied tiers. An SLO violation ($J_\pi > 1$) does not automatically mean hard infeasibility. Only $H_R = H$ varies; H_M, H_C are fixed. The rate unit and compute resource unit are normalized, and migration penalties are dimensionless additive costs rather than seconds of actual downtime. The delay laws are illustrative load-dependent surrogates, not calibrated physical CPU/GPU scheduling or packet-queue models.

2 Full two-service reversal certificate

The following are the exact parameters of the two-service construction in the main paper. Fixed tier capacities are $H_M = 60$, $H_C = 120$; link capacities $(B_1, B_2) = (2.4, 1.8)$; link constants $(\delta_1, \delta_2) = (1, 2)$ and $(a_1, a_2) = (0.5, 0.8)$; radio-side delays $r_A = r_B = 2$. The remaining service parameters are A two-letter

Table 2: Two-service witness parameters: $(b, \eta, c, \tau, p^0_R, p^0_M, p^0_C)$

A	(1, 0.1, 18, 18, 2.5, 2, 1.5)
B	(0.7, 0.5, 12, 20, 2, 1.7, 1.3)

placement string lists the tiers of A and B, respectively. Of the $3^2 = 9$ placements, MM, CM, CC, and CR are strictly worse than MC on their feasible headroom domains, while RC is worse than RM. The remaining normalized costs are

$$\begin{aligned} J_{MC} &= \frac{1133}{2100}, \\ J_{MR} &= \max\left\{\frac{263}{540}, \frac{1241/210 + 2H/(H-12)}{20}\right\}, \\ J_{RM} &= \frac{331/60 + (5/2)H/(H-18)}{18}, \\ J_{RR} &= \frac{2099/390 + (5/2)H/(H-30)}{18}. \end{aligned} \tag{1}$$

The feasible domains of MR, RM, and RR require $H > 12, 18, 30$, respectively. The first envelope crossing is $H_1 = 2460/121 \approx 20.33058$ between MC and MR. The MR cost reaches its constant branch at $H_b = 86892/3461$ with $H_1 < H_b < 78$. The second global crossing is $H_2 = 78$ between that branch and RM. Finally,

$$18(J_{RR} - J_{RM}) = -\frac{7}{52} + \frac{30H}{(H-30)(H-18)} \tag{2}$$

vanishes once for $H > 30$, at the larger root of $7H^2 - 1896H + 3780 = 0$:

$$H_3 = 268.84858\ldots.$$

Exact comparison on the feasible domains therefore gives

$$MC \xrightarrow{H_1} MR \xrightarrow{H_2} RM \xrightarrow{H_3} RR.$$

At $H = 40, 100, 400$ the optima MR, RM, RR are unique, with second-best gaps 0.052487, 0.011179, and 0.002762, respectively. Thus service B follows $R \to M \to R$ as H grows. When only one service exists, MEC and Core costs are constant in H whereas feasible RAN processing cost strictly decreases. Consequently an individual service cannot strictly prefer RAN, then MEC, then RAN at increasing headrooms. Notice that the optimum system cost remains non-increasing, even though the identity of the preferred assignment reverses.

3 Complete five-service exchange certificate

We use the synthetic independent-family witness `n5_14` (population seed 109743). Fixed capacities are $H_M = 60$, $H_C = 120$, $B_1 = 2.4$, $B_2 = 1.8$; delay coefficients match the two-service construction. The operating-cost certificate is evaluated at $H = 30$, with starting placement $s = MCCMR$, the optimum at lower headroom $H = 10$. Every service has radio delay $r_i = 2$. The listed finite-decimal parameters are

Table 3: Exact finite-decimal service inputs for the five-service witness

i	b_i	η_i	c_i	τ_i	p^0_{iR}	p^0_{iM}	p^0_{iC}
1	.4218	.4425	5.8993	17.4387	1.9463	1.7452	1.5212
2	.2379	.3484	4.6096	16.2088	2.1313	1.7235	1.1980
3	.3105	.5310	4.5652	17.1309	2.5887	2.1361	1.5970
4	.3325	.5676	10.7113	18.6267	2.6403	2.1254	1.4458
5	.3973	.2360	4.2146	20.6577	2.0799	1.6023	1.2974

used as *exact rationals* in the certificate. Among $3^5 = 243$ possible placements, 242 satisfy hard capacity constraints at $H = 30$. The best feasible cost at each Hamming distance from the start is All distances 1–3

Table 4: Complete-enumeration minima at each Hamming distance

Distance	Best placement	Minimum J
0	MCCMR	0.5246617927
1	MCCCR	0.5297336756
2	RMCMR	0.5255974337
3	RMMMR	0.5366518597
4	RMMCR	0.5240036549
5	RMMCM	0.5408901965

are strictly worse than the start. A feasible one-service-per-slot sequence exists:

$$MCCMR \to RCCMR \to RMCMR \to RMMMR \to RMMCR.$$

It follows that the minimum number of feasible relocations before reaching a cheaper placement is exactly four. This is stronger than identifying a local optimum, since the complete enumeration checks all states at smaller Hamming distance and the displayed sequence confirms reachability. The first three

Table 5: Operating effects of the feasible four-move sequence

Placement	Link delay (ms)	J	Active SLO service
MCCMR	4.736111	0.524661793	1
RCCMR	4.509106	0.541627375	1
RMCMR	4.204799	0.525597434	2
RMMMR	3.998277	0.536651860	3
RMMCR	4.201771	0.524003655	1

relocations reduce common transport delay, yet each raises the maximum SLO-normalized service delay. The last relocation moves service 4 from MEC to Core, freeing exactly 10.7113 compute units at MEC; despite increased network delay the new operating objective is lower. Migration penalties are not required for this operating-cost trap. The full exact fractional values for all 243 placements are stored in `experiments/multiservice/certificate/n5_14_all_placements.csv`; the quick and deep rational verification methods are described in the reproducibility documentation.

4 Payback definitions, examples, and additional checks

Let the relocation penalty be $\kappa_i = 0.0025$ for each moved service, with a nominal per-slot limit of $m = 1$ (subject to emergency feasibility repair). For a fixed headroom and starting placement s, let $V_K^{\text{out}}(s)$ be the best K-slot accumulated operating-plus-migration cost when the last state is free. Let $V_K^{\text{ret}}(s)$ additionally require that the *last operating-slot placement* equal s. Define

$$K_x = \min\{K \geq 1 : V_K^x(s) < KJ_s(H)\}, \quad x \in \{\text{out}, \text{ret}\}.$$

The diagnostic closed-return constraint is not the true free-terminal orchestration objective and is not the moving-reference condition used by RAR. For a one-move-per-slot sequence $s = \pi^0, \ldots, \pi^r = p$, followed by operation at p through slot $K \geq r$, define

$$g = J_s(H) - J_p(H) > 0,$$
$$B_\gamma = \sum_{j=1}^{r-1}(J_{\pi^j}(H) - J_s(H)) + \sum_{j=1}^{r} C_{\text{rec}}(\pi^{j-1}, \pi^j).$$

The excess cost over staying is

$$C_\gamma(K) - KJ_s = B_\gamma - (K - r + 1)g.$$

For the chosen four-move five-service sequence, $g = 0.000658137822\ldots$ and $B_\gamma = 0.039891290\ldots$. This path changes from a loss at slot 63 to a gain at slot 64. An independently certified rational finite-horizon DP checks that *no other admissible path* improves sooner, and yields $K_{\text{out}} = 64$ and $K_{\text{ret}} = 129$. The

Table 6: Selected witnesses; exact distance and payback diagnostics

	`n3_21`	`n4_11`	`n5_14`
Low to high headroom	$20 \to 50$	$40 \to 70$	$10 \to 30$
Starting placement	RMC	RCMR	MCCMR
Minimum escape moves	3	3	4
Free-terminal payback (slots)	4	12	64
Closed-return payback (slots)	8	25	129
Profitable high pulse (slots)	5	22	128

pulse numbers are minimum durations of the *middle high-headroom segment* in a low–high–low trace with 64 low-headroom slots before and after, not constant-high return diagnostics. Allowing $m = 2$ reduces free-terminal/closed-return pairs to 2/5, 6/13, and 18/37, respectively. Allowing all five services to move together still leaves a 16-slot constant-high payback for the direct five-service exchange. These selected values show that relocation distance, number of distinct services moved by an optimal plan, and accumulated return-adjusted payback are different quantities.

5 Synthetic populations, ablation, and controller traces

The primary experiment retains the identical three-tier, two-link model. It draws 32 ($n = 3$), 24 ($n = 4$), and 16 ($n = 5$) synthetic populations, each with total input rate 1.7 and total compute demand 30. Half the population instances use independent heterogeneity and half a compression–compute trade-off. The complete generation ranges, seeds, code, and data are in the public release. No population is filtered by outcome. These are counts of populations containing at least one sampled witness on the predefined

Table 7: Original structural occurrence breakdown

n	3	4	5
Populations	32	24	16
Sampled reversal	23	18	14
Inherited strict local trap	32	22	16

headroom grid, not deployed-network prevalence rates. A local-trap condition is checked separately from reachability. Setting all $\eta_i = 1$ also drives aggregate downstream traffic to 1.7 against link capacity 1.8, so mechanism ablations do not isolate small perturbations or imply generic necessity.

Table 8: Paired ablation counts, each out of the same 72 populations

Variant	Reversal	Trap
Full model	55	70
No traffic transformation ($\eta_i = 1$)	0	4
Load-independent transport delay	1	5
Load-independent compute delay, hard capacity retained	53	10

Each population is exposed to six predefined headroom traces: pulse, two ramps, oscillation, abrupt withdrawal, and seeded blocks, for 432 primary matched traces. Under nominal $m = 1$, $\kappa_i = 0.0025$, emergency repair exceeds the budget in 212 myopic transitions and 197 each for MPC64 and RAR-A64. Every controller knows the current headroom; each has exact forecasts only within the windows it queries, and all know the experimental trace endpoint. Offline DP uses full-trace knowledge. The mean gap below is the simple mean across *traces* of the percentage total-cost gap against offline, rather than a pooled per-slot physical-latency improvement. In these 432 primary traces, myopic is suboptimal on 380; MPC64 matches

Table 9: Primary-trace controller mean percentage gaps

Controller	$n = 3$ (192)	$n = 4$ (144)	$n = 5$ (96)
Myopic	1.45175	0.68858	1.64230
MPC8	0.00823	0.00758	0.00691
RAR8	0.05908	0.06442	0.05835
MPC64	0	0	0
RAR-A64	0.00035	0	0.00023

offline in all 432 and RAR-A64 in 428. At preview eight, RAR8 beats MPC8 on 18, ties on 198, and loses on 216; MPC8 has four cases worse than myopic, and RAR8 none. This does not prove RAR superiority over MPC or universal sufficiency of any fixed preview.

The two pulse illustrations in the conference paper are *selected sensitivity examples separate from the 432 primary traces*. The $n = 4$ 24-slot pulse allows a profitable four-service exchange that earns 0.009951 versus myopic and is identified by MPC64 and RAR-A64, but not by eight-slot controllers. For the selected $n = 5$ 96-slot pulse, MPC64 incurs 0.020550 extra accumulated cost (0.020000 migration and 0.000550 operating) compared with myopic. In a separate 256-slot constant-$H = 30$ experiment, MPC64 and offline save 0.126618 versus myopic but RAR-A64 only 0.000255, until the longer $K = 129$ window is available.

A third, separate sensitivity grid contains 270 combinations (three selected witnesses, 15 pulse durations, six migration-penalty values). Offline improves on myopic in 94; RAR-A64 misses all profitable savings in six such cases but avoids losses relative to myopic in this grid, while MPC64 loses to myopic in six cases. In the primary suite myopic records 50 slots with $J > 1$ out of 46,080 tested operating slots, compared with zero for the tested preview policies; this is only an observed diagnostic.

Finally, we independently perturb the three selected witnesses by up to $\pm 2\%$ in 50 unfiltered trials each, renormalizing total load. For each perturbed system, the low-headroom optimum is *recomputed*. A strict higher-headroom trap persists in 48/50, 33/50 and 25/50 cases; the four-move five-service obstruction persists in 11/50. The *original* starting placement remains unchanged in 25/50, 31/50 and 43/50 cases, respectively. These quantities characterize persistence of the phenomenon under the stated reoptimization protocol, not stability of an identical initial placement in all trials.

6 RAR baseline condition and proof details

Let a feasible, deterministic shadow-myopic trajectory $\bar{\pi}_0^{\text{myo}}, \ldots, \bar{\pi}_T^{\text{myo}}$ begin in the same placement as the RAR controller. A retained, unexecuted incumbent plan is completed before the fallback follows the shadow reference. An alternative must cover the whole incumbent and, for a comparison window ending before T, end at the *same reference placement* as the fallback. Only a strictly cheaper full-window plan is accepted. At T, the endpoint is free.

Under correct forecasts in each queried window, the same feasibility and migration-cost accounting, and the retained-incumbent rule, an accepted plan replaces a prospective full trajectory by one of cost smaller by $\Delta_k > 0$. The plans agree beyond the comparison endpoint, so all later actions can be shared. Induction preserves a feasible full continuation, and telescoping gives $C_{\text{myopic}} - C_{\text{RAR}} = \sum_{k=1}^{T} \Delta_k \geq 0$. Since offline DP minimizes over the feasible full trajectories, $C_{\text{offline}} \leq C_{\text{RAR}} \leq C_{\text{myopic}}$. This is a conditional use of the

known feasible-baseline or fortified rollout principle, not a new general theorem. It gives no matched-preview MPC dominance or forecast-error robustness.

7 Evidence files, licenses, and versioned publication

The versioned public research artifact is maintained at

`https://github.com/jozsefbirotmit/edge-orch-reprod`.

The first archived version has the version-specific identifier `https://doi.org/10.5281/zenodo.24200109` (v1.0.0). This v2.4 supplement updates that previously published companion document; its next archival DOI must be copied from Zenodo only *after* a new tagged release has been ingested. Never substitute the older version DOI for a revised release.

The repository contains both the original exact-placement and multiservice experimental archive and the ablation study in a single stable directory tree. Principal files include:

- `experiments/multiservice/verify_n5_certificate.py` — exact witness validation and payback checks;
- `experiments/multiservice/certificate/n5_14_all_placements.csv` — all placement costs;
- `experiments/multiservice/data/fraction_certificates.json` — independent rational certificates;
- `experiments/ablation/verify_results.py` — mechanism ablation verification;
- `tools/verify_release.py` — archived release integrity and smoke tests;
- `SHA256SUMS.txt` — full release file integrity manifest.

The code and executable scripts are provided under the MIT License. Original synthetic data, figures, and written documentation are provided under CC BY 4.0; consult `LICENSE`, `LICENSE-DATA.md`, and `docs/LICENSING.md` for coverage. The corresponding author has confirmed Nokia/BME dissemination approvals and the two-author copyright attribution.

For the release-root verification suite, run:

```
python -B tools/verify_release.py --quick
```

For the independent deep payback verification, run from the release root:

```
python -B experiments/multiservice/verify_n5_certificate.py --deep-payback
```

The underlying figure PDFs and numerical data are archived. A fully automated recreation of the exact conference manuscript Figures 2 and 3 has not been independently certified, and the simulations use synthetic (not measured telecom) workloads. These limitations do not affect the stated exact rational certificates.